

Python - The basic!



Lesson 0: Hello World! - Your First Python Program

Running your first code

```
[1]: print("Hello world")
```

Hello world

Comments

```
# Display "Hello world" text  
print("Hello world")
```



Data Types and Variables

Data types

Data Types

In programming, data types are used to tell computer the type of data

Python has a small set of built-in types storing different type of data:

- **Integers (int)**: Whole numbers (e.g., 10, -5).
- **Floats (float)**: Decimal numbers (e.g., 3.14, -12.5).
- **Strings (str)**: Text enclosed in quotes (e.g., "Hello", 'This is a string').
- **Booleans (bool)**: True or False values.
- **None**: empty data

*Note: these built-in data type are called **primitive/scalar data type***



1. Integer

```
[5]: 1
```

```
[5]: 1
```

```
[9]: type(1)
```

```
[9]: int
```

2. Float

```
: 1.0
```

```
: 1.0
```

```
: type(1.0)
```

```
: float
```

3. Boolean

True

True

False

False

`type(False)`

bool

4. String

```
"True"
```

```
'True'
```

```
'True'
```

```
"""  
True  
False  
"""
```

```
'\nTrue\nFalse\n'
```

```
print("""  
True  
False  
""")
```

```
True  
False
```

```
>>> type("True")
```

```
>>> str
```

```
>>> type("""  
True  
False  
""")
```

Problems with Floating Point

Floating-point numbers are represented in computer hardware as base-2 (binary) fractions, which can lead to unexpected precision issues.

```
# Example of precision loss in Python
```

```
>>> 3.3 * 33
```

```
108.89999999999999
```

For more details, see the official Python tutorial:

<https://docs.python.org/3/tutorial/floatingpoint.html>

Key Takeaway: Be mindful of precision when performing financial or scientific calculations.

Values and Types

Strings: `'2'` and `'42.0'` are strings because they are enclosed in quotation marks.

Number Separators: Numbers can be separated by underscores (`_`) for readability, e.g., `1_000_000`.

Legal Values: ~~`1,000,000`~~ is not a legal integer in Python, though it may be a legal value in other contexts.

Variables

What is Variables?

Variables are containers for storing data values.

You can create a variable simply by assigning a value to it using the **assignment operator (=)**.

```
age = 30 # Integer  
pi = 3.14159 # Float  
name = "Alice" # String  
is_student = True # Boolean
```

```
type(age)
```

```
int
```

```
age
```

```
30
```

Variable name and Identifiers

A Python identifier is a name used to **identify a variable, function, class, module or other object**.

Starts with a letter A-Z or a-z

Example: `my_var`, `MY_var`, ~~`1Var`~~

Starts with an underscore (_)

Followed by zero or more letters, underscores and digits (0-9).

Example: `_MY_VAR`, `_VAR1`, `__var`, ~~`_1MY_VAR`~~

Does NOT allow punctuation characters

Such as `@`, `$`, `%` within identifiers.

Example: ~~`$my_var`~~, ~~`@var`~~, ~~`&my_var`~~

Variable name and Identifiers

- Does **not** allow **Python's keywords**
 - Example: ~~class~~ = ~~'Class A'~~
- **Python 3 has these keywords**

False	class	finally	is	return
None	continue	for	lambda	try
True	def	from	nonlocal	while
and	del	global	not	with
as	elif	if	or	yield
assert	else	import	pass	
break	except	in	raise	

Data type conversion

Data type conversion

In order to convert between data type, we can be done by casting type with built-in function below:

- `int()`
- `float()`
- `bool()`
- `str()`



String or Boolean to int or float

```
: (int(False), float(True))
```

```
: (0, 1.0)
```

```
: float('1000.22')
```

```
: 1000.22
```

```
: int('1000')
```

```
: 1000
```

String or number to bool

```
(bool('False'), bool('True'))
```

(True, True)

```
bool(1), bool(0)
```

(True, False)

Int, float or boolean to string

```
|: str(1)
```

```
|: '1'
```

```
|: str(False)
```

```
|: 'False'
```

```
|: str(1.0001)
```

```
|: '1.0001'
```

Practices

Practices

- Open Jupyter Notebook
- Create new Notebook
- Declare 2 variable with value of "12.1", and 10
- Convert them to number and sum them

